

Question: Define $p = \sum_{k=1}^{\infty} \frac{1}{k^2}$ and $q = \sum_{k=1}^{\infty} \frac{1}{k^3}$. Find a way to write $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q .

Answer:

SABER

<think> Okay, so I need to find a way to express the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q , where $p = \sum_{k=1}^{\infty} \frac{1}{k^2}$ and $q = \sum_{k=1}^{\infty} \frac{1}{k^3}$. Let me start by understanding what this double sum represents and how I can relate it to p and q . First, let me recall that p is the famous Basel problem, which converges to $\frac{\pi^2}{6}$, and q is the Riemann zeta function at 3,

<omitted>

So: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = \sum_{n=2}^{\infty} \frac{1}{n^3} (n-1)$

SC: 1.0 CS: 0.83 RSS: 0.88

, let me check that. If I fix $n = j + k$, then for each n , there are $n - 1$ pairs (j, k) , so the double sum becomes: $\sum_{n=2}^{\infty} \frac{1}{n^3} (n-1)$

<omitted>

Similarly, $q = \sum_{n=1}^{\infty} \frac{1}{n^3}$, so $\sum_{n=2}^{\infty} \frac{1}{n^3} = q - \frac{1}{1^3} = q - 1$. Therefore, substituting back: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = (p-1) - (q-1) = p - q$

SC: 1.0 CS: 0.99 RSS: 0.99>0.95

</think>

To evaluate the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$,

<omitted>

Substituting these into the expression: $\sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n^3} \right) = (p-1) - (q-1) = p - q$

Answer: $\boxed{p - q}$

Token number: 1626

Vanilla

<think> Okay, so I need to find a way to express the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$ in terms of p and q , where $p = \sum_{k=1}^{\infty} \frac{1}{k^2}$ and $q = \sum_{k=1}^{\infty} \frac{1}{k^3}$. Hmm, let me think about how to approach this. First, I remember that double sums can sometimes be simplified by changing variables or by recognizing patterns.

<omitted>

Therefore, the sums from $n=2$ to infinity are just $p - 1$ and $q - 1$, respectively. So substituting: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = (p-1) - (q-1) = p - q$. Wait, so does that mean the original double sum is equal to $p - q$? Let me verify this because that seems straightforward, but I want to make sure I didn't make a mistake in the substitution.

Let me check with small values. Let's compute partial sums numerically. Let's take the original double sum up to $j=1$, $k=1$, then $j=1$, $k=2$, etc., and see if it aligns with $p - q$. But maybe that's too tedious. Alternatively, let me check the steps again.

Original double sum: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$. Changing variables to $n = j + k$. For each $n \geq 2$, the number of pairs (j, k) with $j + k = n$ is $n - 1$. Therefore, $\sum_{n=2}^{\infty} \frac{1}{n^3} (n-1)$. Then splitting into $\frac{1}{n^2} - \frac{1}{n^3}$. Then sum from $n=2$ to ∞ of $\frac{1}{n^2}$ is $p - 1$, same with sum $\frac{1}{n^3}$ is $q - 1$. Therefore, total is $(p-1) - (q-1) = p - q$.

<omitted>

Yes, that's straightforward. Therefore, this seems to confirm again that the answer is $p - q$. Therefore, I think my initial conclusion is correct. Therefore, the answer is $\boxed{p - q}$.

</think>

To evaluate the double sum $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3}$,

<omitted>

Substituting back, we have: $\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{(j+k)^3} = (p-1) - (q-1) = p - q$

Final Answer: $\boxed{p - q}$

Token number: 5840